# DeepFake Analysis Report

Explainable Deepfake Detection — Xception-LSTM with LIME · Research prototype

| | |
|---|---|
| Analysis ID | 3495ed1b-79ae-449a-b7cf-1b9c3ce4232f |
| Generated | 2026-11-02 08:31 UTC |
| Analyzed at | 2026-11-02T08:25:47.156934+00:00 |
| Status | COMPLETED |

## 1. Executive summary

| FAKE | raw fake score 0.78528 | decision threshold 0.6 (dev-v1) | model version v2 |
|---|---|---|---|

fake_score 0.78528 vs decision threshold 0.6 (dev-v1)

**This score is an uncalibrated sigmoid output — it is not a probability and not forensic proof.**

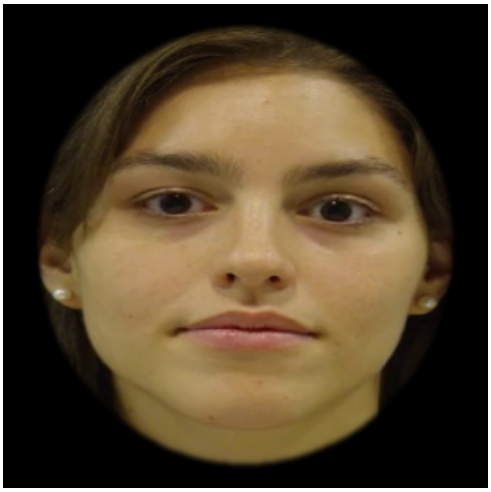
## 2. Media information

| | |
|---|---|
| Filename | test_face.jpg |
| Media type | image |
| File size | 211.6 KB |
| Resolution | — |
| Duration | — |
| FPS | — |
| SHA-256 | a8fc6663ae969e20b38cbe0f7f0d086c33417e76539685ed362c85f31c643c61 |
| Frames extracted / analyzed | 1 / 1 |

## 3. Prediction analysis

| | |
|---|---|
| Raw model output (fake score) | 0.78528 |
| Decision method | Replicated-frame compatibility mode (video model applied to a single face crop) |
| Decision threshold | 0.6 (dev-v1) |
| Score calibrated? | No — sigmoid output, uncalibrated |
| Model version | v2 |
| Window score variation | — (no window scores) |

## 5. Visual evidence



Processed face (elliptical mask applied)

LIME was not requested or is unavailable for this analysis.

LIME highlights regions of a single replicated frame that most influenced the model's response. It does NOT prove those pixels were manipulated and does not explain the temporal (LSTM) pathway.

## 6. Technical details

| | |
|---|---|
| Model | Xception-LSTM · version v2 |
| Checkpoint SHA-256 | 9ee14530166b842ce0aa6ac74500ac138993f5fbe82a927df7ddee0ffc261228 |
| Sequence length | 32 frames @ 299×299 px |
| Preprocessing | OpenCV DNN face detection (confidence 0.5) → 20% margin crop → 299×299 resize → elliptical background mask → [0, 1] normalization |
| Prediction method | Replicated-frame compatibility mode (video model applied to a single face crop) |
| Processing duration | 10.9 s |

## 7. Interpretation

The model scored this media above the configured decision threshold, which maps to its FAKE class. The score exceeds the threshold by 0.185. Scoring method: replicated_frame_sequence. Because the model is uncalibrated and was evaluated as weak on held-out data (see evaluation/EVALUATION.md), this assessment must be treated as a research-grade indication and reviewed by a human before any consequential use. Quality warnings were raised during processing.

## 8. Warnings

• Image analysis reuses the VIDEO model by replicating one face crop across the frame axis; it is not a separately validated image classifier.

## 9. Research limitations

1. Scores are sigmoid outputs from an uncalibrated model; they are not proven probabilities.

2. Models were trained only on the DeepFake Detection (DFD) subset of FaceForensics++; performance on other manipulation methods is unknown.

3. The train/val split has identity leakage, so measured metrics are optimistic.

4. Only the most confident face per frame is analyzed; multi-face videos are not fully covered.

5. LIME explains a replicated static frame; it does not attribute the temporal pathway.

6. Face-detection variability across OpenCV builds can change scores for the same video.